

Emergence of Symbolic Language from Perception via Physical Symmetry

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Abstract

A central question in artificial intelligence and cognitive science is how symbolic structure arises from perceptual experience. In this work, we show that discrete symbols and their contextual relations can emerge in neural networks when the self-supervised learning process is guided by intrinsic physical symmetry. Without predefined labels, our model trained only on grouped visual scenes can acquire a generalizable numerical language that supports addition beyond its training experience and transfers across novel visual forms. The same mechanism also enables a model to discover pitches and scale relations from raw audio alone. Together, our findings identify intrinsic physical symmetry as a generative principle that organizes perception into symbolic language.

Keywords: symbolic language, intrinsic physical symmetry, self-supervised learning, vector quantization

1 Introduction

Symbolic language underpins rationality, science, and human civilization, enabling us to think and coordinate far beyond the limits of direct perception. Yet, a deep mystery, central to both cognitive science and artificial intelligence (AI) remains: how can symbolic structure emerge from raw perceptual experience [1, 2]? Early AI systems were explicitly symbolic yet perceptually blind; modern neural networks, in contrast, perceive richly but lack self-generated symbolic structure [3]. Bridging the two—allowing

symbolic reasoning (System 2) to emerge autonomously from perception (System 1)—would mark a major step toward understanding intelligence itself [4–6].

Among various symbolic constructs, natural numbers and addition form a uniquely foundational core: numbers provide a vocabulary of discrete symbols, while addition captures the fundamental relations among them. Children grasp numbers intuitively [7, 8]; formal mathematics builds on them rigorously. In this article, we ask: can a learning system, exposed only to grouped visual scenes (e.g., one, two, three objects as shown in Fig. 1A), spontaneously discover the concepts of number and addition without any symbolic labels or counting supervision?

We discovered that a self-supervised neural network, when equipped with physical symmetry on its latent structure, can spontaneously form discrete representations that behave like natural numbers and support addition (as shown in Fig. 1D). As a highlighted result, trained on only 30% of possible visual combinations, the model learns a symbolic language that generalizes to unseen addition questions with 96% accuracy, whereas the same network trained without symmetry reached only 24%, only memorizing without generalization capability.

Importantly, the model contains no predefined numerical structures or group priors. Its symbolic behavior arises purely from intrinsic symmetry—an organizing principle that shapes the learned abstractions and corresponding context dependency rather than the raw input space. In the following sections, we show that the emergent symbolic structure not only resembles natural numbers, but also supports inverse operation (subtraction) and even adding fractions that were never seen during training, all without symbolic instruction. Furthermore, the same principle extends to another domain—music—where the model discovers pitch and scale from raw audio, suggesting a broader capacity to extract symbolic languages from perception through symmetry-driven learning.

Beyond a technical advance, our work proposes a new paradigm: symbolic language, like physical law, may emerge from the symmetry of structured interaction. This points to a deeper unity between mind and matter: features across both physical and cognitive domains may be governed by shared principles of symmetry [9]. In essence, symmetry is not merely a constraint of nature—it can serve as a generative principle for abstraction itself.

2 Results

2.1 The Emergence of Numbers and Addition

As a first test of whether intrinsic physical symmetry can stabilize an emergent symbolic language, we began with the simplest symbolic domain: small natural numbers and addition. This task has a clear developmental analogue: children can learn numerical concepts from simple visual groupings and generalize from a few examples to unseen addition problems [10–13]. In the corresponding self-supervised setting, we exposed a neural network only to grouped visual scenes without symbolic labels. The model must map visual inputs to abstract number symbols, compose two symbols to represent their sum, and generalize to held-out addend pairs not observed during training (Fig. 1A–C).

Our dataset contains a collection of image triplets, each representing two natural numbers (addends) and their sum, with the maximum number bounded by twenty. The dataset has three versions, progressively introducing more challenges in abstracting numbers from visual appearances: 1) dot arrays (Fig. 1A), 2) more patterns with diverse colors, blur, or shapes (Fig. 1B), and 3) realistic icons with random positions and sizes (Fig. 1C). In this section, we focus on the results of version 1: learning a symbolic language from dot arrays. We provide results for version 2 in Supplementary Section S.1.7, which yield very similar findings. We will also show that models trained on version 1 can be adapted to version 3 to recognize realistic patterns in the next section.

Emergent Language from Perception with Intrinsic Physical Symmetry (ELPIS) learns numerical symbols and their additive relations from such images through three components (Fig. 1D). First, a vector-quantized variational autoencoder (VQ-VAE) maps each input image to a discrete latent code from a fixed-size learned dictionary (Fig. 1D-i) [14]. This learned dictionary provides the basis for symbol formation: each perceptual input is compressed into a single discrete code, yielding a set of reusable symbolic units. However, these codes remain semantically unstructured, as no constraints are imposed on their relations. Second, a neural reasoning module takes the symbolic codes corresponding to two inputs and predicts the code of their sum (Fig. 1D-ii). In this way, the model learns relations among the symbols directly from visual observations, but its generalization remains limited, as it tends to memorize relations encountered during training. Finally, intrinsic physical symmetry (Fig. 1D-iii)—the key mechanism enabling generalization—is enforced directly at the level of latent representations through the reasoning function itself. This constraint organizes the symbolic space by enforcing consistency across alternative computational paths, allowing the model to infer unseen relations. **In summary**, *VQ provides discrete tokens, the reasoning module learns their observed relations, and intrinsic symmetry organizes these relations into a generalizable symbolic system.*

Physical symmetry is characterized by a transformation T and a law R that commute, i.e., $T(R(\cdot)) = R(T(\cdot))$. A classic example is that Newtonian dynamics are invariant under Galilean transformations. In machine learning, we reinterpret this principle in latent space: R corresponds to a learned reasoning function over symbolic codes, while T represents a transformation within the same latent space. Unlike prior approaches that assume predefined transformation groups (e.g., translations or rotations) [15], here the *transformations need to be learned from data*, as the symbolic language itself is unknown. In other words, the transformation T is not predefined but learned jointly with R , as both describe contextual relations in the latent representation space. We refer to this learning principle as *intrinsic physical symmetry*.

The general condition $TR = RT$ can manifest in different forms and parameterizations depending on the structure of R , without requiring a commutative reasoning function (see Supplementary Sections S.1.2 and S.1.9). For the addition task, where the reasoning function takes two inputs and produces one output, we can define $R_b(a) = h_\theta(a, b)$, where a and b are learned symbolic codes. Similarly, in this instantiation, the transformation $T_c(a)$ can be expressed using $h_\theta(a, c)$. The symmetry condition $T_c(R_b(a)) = R_b(T_c(a))$ then requires $h_\theta(h_\theta(a, b), c) = h_\theta(h_\theta(a, c), b)$ for

arbitrary learned codes a , b , and c . During learning, this constraint plays two roles. First, it improves sample efficiency by introducing additional computational paths that must agree, effectively augmenting supervision in the latent space. Second, it enhances generalization by applying the reasoning function h_θ to novel, unseen symbol combinations, enabling the model to reason about hypothetical compositions—akin to analogy-making in human cognition.

Notably, a structure analogous to associativity in the learned representation space arises as a consequence of this symmetry constraint, rather than being imposed a priori. As we show later, intrinsic physical symmetry extends beyond arithmetic as a meta-level principle. In the musical domain, although R and T take different forms, the underlying idea remains the same: both transformations and relations are learned jointly from data.

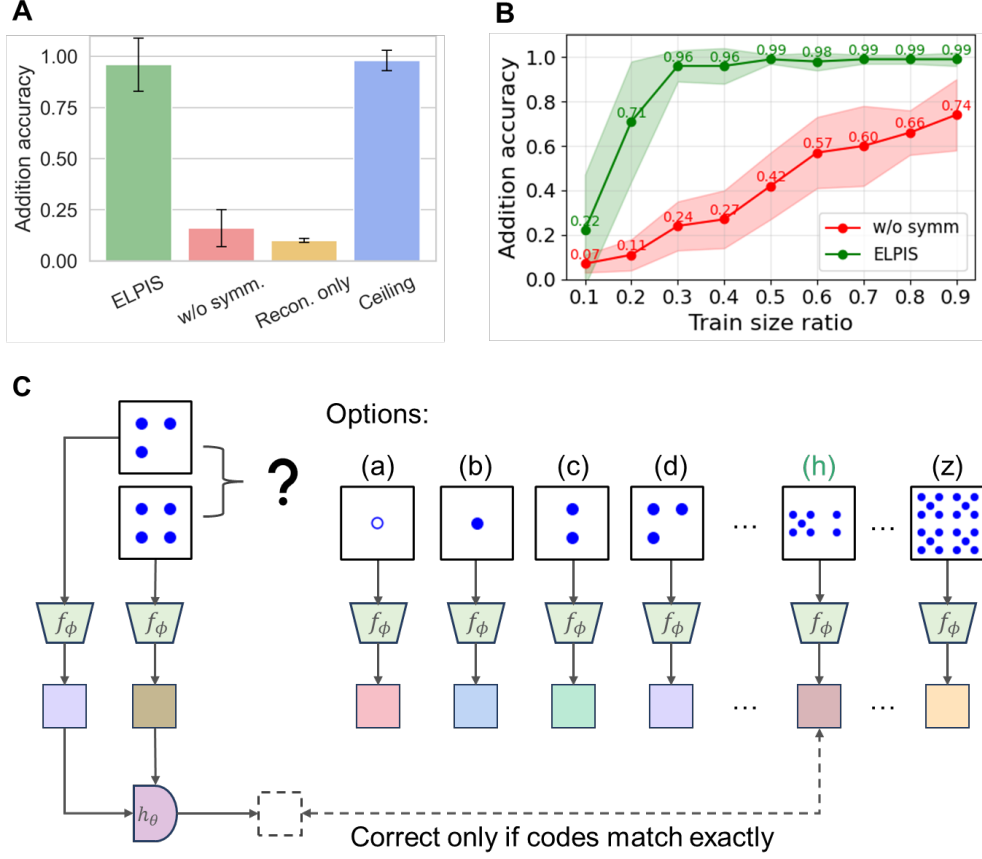


Fig. 2 The evaluation of the learned symbolic language: generalizability of “addition” (reasoning) on unseen questions. (A) A comparison of addition accuracy on unseen addend pairs between ELPIS and alternative models. (B) Addition accuracy as a function of training-data ratio, comparing ELPIS with the model without symmetry: ELPIS rapidly achieves near-perfect accuracy with a small fraction of training pairs, whereas the w/o symmetry model improves gradually with increasing data, revealing a pronounced gap in sample efficiency and generalization. (C) Schematic illustration of the hard-matching criterion. For each query (x_m, x_n) , the predicted symbolic code for the sum is regarded as correct only if it matches the exact code of the sum.

The main evaluation focuses on whether the symbolic language learned by ELPIS can *generalize* to unseen addition problems when trained only on a subset of image triplets. The addition accuracy is computed using the **Hard Matching** criterion, which requires the reasoning result to match the exact code of the correct sum. (A complementary distance-based criterion is described in Methods and Supplementary Section S.1.5.1 and yielded a very similar pattern.) A highlighted result is shown in Fig. 2A: trained on only 30% addend pairs, ELPIS achieves around 96% accuracy (green bar) on the held-out test set, approaching the performance ceiling under full-data training (blue bar). In contrast, removing intrinsic symmetry (w/o symm.) leads to a dramatic performance degradation, with accuracies of 16%. Further ablating the

reasoning module reduces the model to a plain VQ-VAE trained only for reconstruction (Recon. only), resulting in near chance-level performance. We further examine generalization capability as a function of training set size (Fig. 2B). ELPIS reaches near-perfect accuracy with less data and saturates rapidly. By contrast, without intrinsic symmetry, generalization improves only gradually as more data is added, remaining far from saturation even when 90% of addend pairs are observed. Together, these results demonstrate the strong ability of ELPIS to perform unseen addition problems, suggesting that intrinsic symmetry enables generalization beyond limited experience or mere memorization [16].

Besides strong generalizability in the addition task, we find that the symbols learned by ELPIS exhibit *structural properties* characteristic of natural number representations. We analyze this through interpolation accuracy, latent-space orderliness, and transfer-learning to the subtraction task (Fig. 3). None of these properties was explicitly trained; they emerge spontaneously from the learned representations.

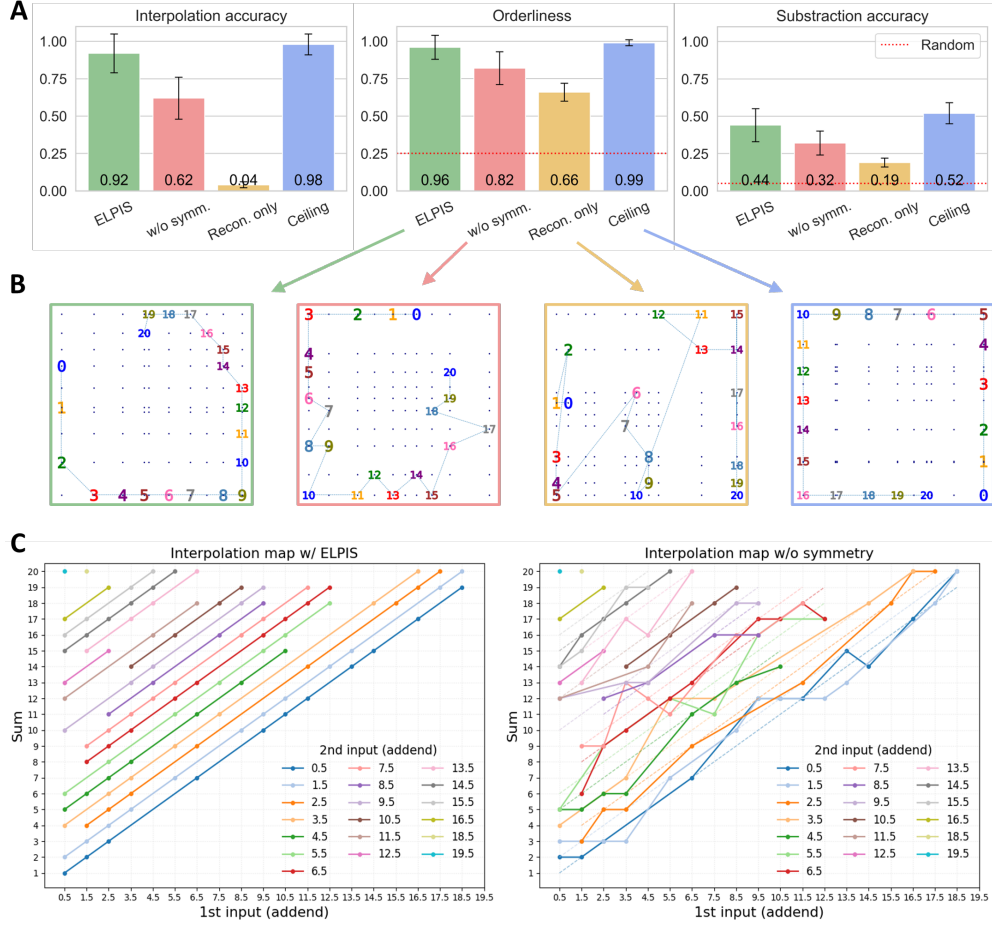


Fig. 3 Further analyses on the representation spaces (of symbolic natural numbers) learned by different models. Panel A reports aggregate quantitative summaries, whereas panels B and C show representative qualitative visualizations. **(A)** Overall analysis results of different models on three metrics: 1) interpolation accuracy, 2) orderliness, and 3) subtraction accuracy. Interpolation accuracy measures whether the learned integer representation and addition can generalize to fractional numbers. Orderliness measures how orderly the learned symbolic representations are arranged according to their numerical semantics. Subtraction accuracy evaluates how well the learned symbolic representations from addition are reusable in subtraction. **(B)** A 2D visualization of the latent spaces: highlighting that the symbols learned by ELPIS are arranged in a more orderly manner in the latent space (i.e., with high “orderliness”), closer to the ideal (ceiling) scenario. **(C)** Detailed interpolation maps for ELPIS (left) and for the alternative model (right, which is without symmetry), highlighting that ELPIS achieves semantically coherent, near-perfect interpolation accuracy, a property lost when symmetry is missing. The x-axis denotes one interpolated addend, and the colored lines denote the other addends. The y-axis shows the predicted sum. In this case, we only tested whether $n.5 + m.5$ matches $m + n + 1$, where the latent code of $x.5$ is obtained by interpolating the codes of x and $x + 1$. To make a fair comparison, we create interpolation cases only adjacent to training pairs (e.g., we test $3.5 + 5.5 = 9$ only if $3 + 6$ or $4 + 5$ appears in the training set). For panels A–C, all models except the ceiling model are trained on the same 30% subset as in Fig. 2A; the ceiling model no-symmetry objective but is trained on all addend pairs. The random baseline is evaluated on symbolic representations randomly arranged in the latent space.

Interpolation analysis reveals that the latent space encodes meaningful semantics beyond integers. Although the model is trained only on integers, it supports coherent reasoning over intermediate values. Concretely, we construct interpolated symbols by taking midpoints between adjacent integers (e.g., between 3 and 4 to form a midpoint code for 3.5; see Methods), and query the learned reasoning function on these simple fractional inputs. Fig. 3A shows that ELPIS achieves high interpolation accuracy (92%), substantially outperforming the model trained without symmetry (62%) and the reconstruction-only baseline (4%), approaching the ceiling (98%). This advantage is illustrated qualitatively in Fig. 3C, where the x-axis denotes one interpolated addend, the colored lines correspond to the other addend, and the y-axis shows the predicted sum: ELPIS exhibits near-linear trajectories consistent with the arithmetic structure, whereas removing symmetry produces distorted interpolation behavior.

Symmetry also induces a highly ordered latent structure resembling a mental number representation [17, 18], as shown in Fig. 3B. The background plane visualizes the learned latent space, where each point corresponds to a discrete code and distances reflect latent similarity. **Orderliness** is quantified by whether adjacent integers are embedded proximally, with an ideal score of 1.0 (see Methods). As summarized in Fig. 3A, ELPIS achieves higher orderliness (0.96), better than the model without symmetry (0.82) or with reconstruction only (0.66), and close to the ceiling (0.99). Fig. 3B provides a representative visualization: ELPIS organizes symbols into a near-monotonic arrangement close to the ideal (ceiling) scenario, while alternative models exhibit noticeable local violations.

Finally, we evaluate whether the learned symbols support reuse beyond the training task (see Methods). We freeze the encoder and symbolic codes and train a new reasoning module for the subtraction task with limited supervision. Fig. 3A shows that representations learned by ELPIS achieve substantially higher subtraction accuracy (44%) than those learned without symmetry (32%) or with reconstruction only (19%), consistently exceeding a random baseline and approaching the ceiling (52%).

Together, these results indicate that intrinsic physical symmetry not only improves generalizability for the addition task but also induces a human-aligned representational space of numbers, manifesting coherent interpolation, ordered geometry, and even transferable algebraic utility.

2.2 Generalizing to Complex Visual Scenes

One may ask whether abstract number concepts learned from simple patterns can generalize to more complex visual scenes also without relying on number labels. If so, the model would be able to count and perform addition on more realistic scenarios in a fully self-supervised manner.

To this end, we construct a new dataset that contains images of more complex and realistic icons, with randomized sizes and non-overlapping positions, as shown in Fig. 1C (see Methods and Supplementary Section S.1.8). To perceive these new patterns, we adopt DINO ViT [19], a popular pretrained visual encoder that captures rich representations through self-supervised learning. However, such a visual encoder alone lacks the abstraction power to explicitly align inner representations to numerical concepts [20, 21]. Hence, we introduce a lightweight MLP adaptor v_ω that projects

these representations into the existing numerical concept space—symbols previously emerged from simple dot patterns. Specifically, we partition each image into two complementary parts recursively by masking a subset of icons and, during adaptation, we enforce the numbers encoded from the two parts to be mapped (by the existing reasoning function $h_{\theta'}$) to the corresponding number of the full image, as shown in Fig. 4A.

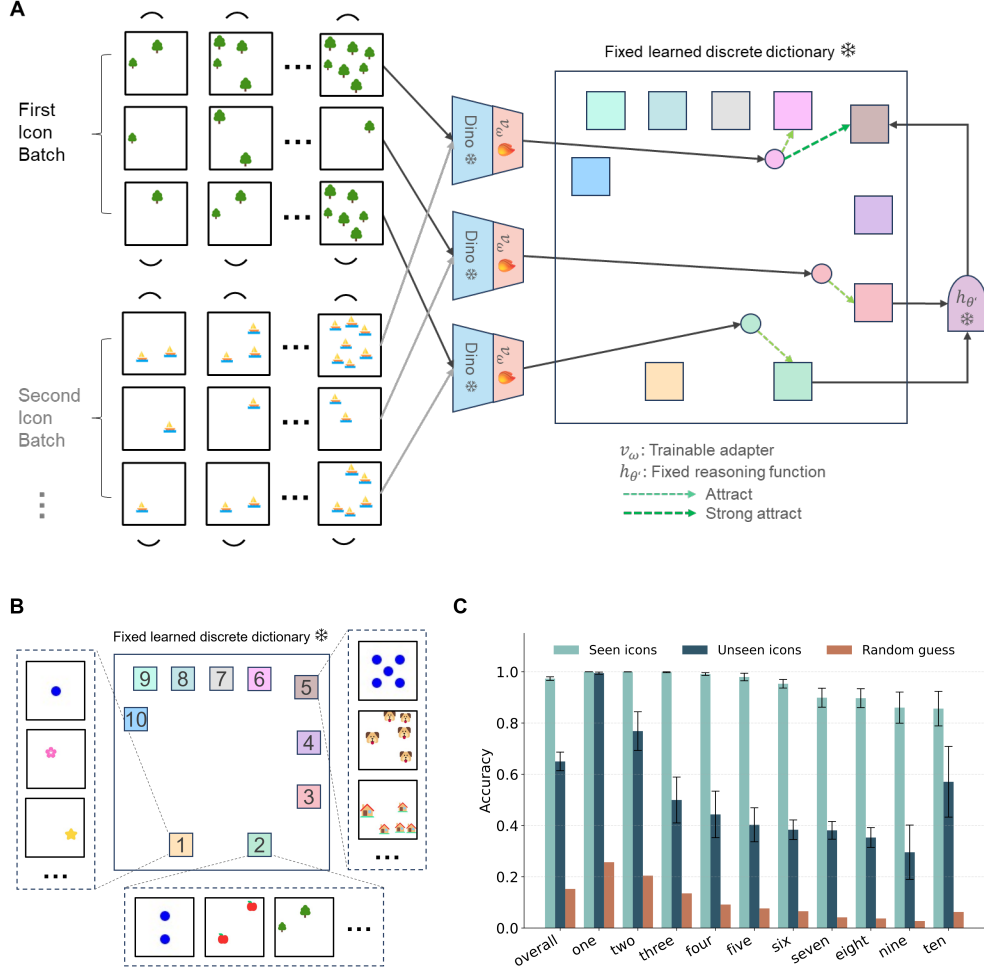


Fig. 4 Main results on counting accuracy for more complex and realistic iconic patterns after perceptual adaptation. (A) An illustration of our perceptual adaptation process, which enables ELPIS to further process (recognize their numbers and perform addition on) more realistic iconic patterns in a self-supervised manner. Training data are triplets of iconic patterns, each containing *randomly sized* and *positioned* objects drawn from 10 icon types. The adaptation framework includes three components: a fixed ELPIS codebook and reasoning function $h_{\theta'}$, a fixed DINO ViT encoder, and a trainable MLP adapter v_ω . For stability, we restrict the ELPIS dictionary to symbols 1 – 10, resulting in quantization over a smaller codebook and therefore a slightly different parameterization of $h_{\theta'}$. Correspondingly, the training data are restricted to 1 – 10 objects. (B) An illustration of different types of iconic patterns are mapped to the correct number representations after adaptation. (C) Per-number recognition accuracy on both seen and unseen icons: the number representation and addition function learned by ELPIS generalizes successfully to random size, location, and unseen icons.

Experimental results show effective adaptation: iconic patterns can be mapped to the same numerical symbols as dot patterns (Fig. 4B). Quantitatively, Fig. 4C reports counting accuracies for numbers 1 – 10 in the 10 of 20 independent runs that converged according to a training-stage criterion (Supplementary Table S2). On icons seen during adaptation, these converged runs achieve consistently high performance across numbers, with an average accuracy of 97% and only a mild degradation as count increases. Under a more challenging condition—testing on iconic objects never presented during adaptation—the model retains substantial performance above random guessing, achieving 65% average accuracy.

Above all, these results show that ELPIS learns a symbolic language not tied to specific visual appearances. By adopting a powerful perception module and training only a small adapter, it can be applied to more complex visual scenarios, preserving both countability and additive compositionality. This mirrors human cognition—numerical concepts acquired in simplified settings can be reused in diverse and unencountered environments [22, 23].

2.3 Symbolic Language Emergence Beyond Math

We have seen that by observing images alone without symbolic labels, ELPIS can learn numerical concepts and addition under intrinsic symmetry constraints. A natural question follows: would the same epistemic principle support the emergence of symbolic language in another domain? To this end, we extend our framework to the domain of sound and music. Like mathematics, music provides a fundamental form of abstraction: discrete pitches act as symbols analogous to natural numbers, while scales and keys impose structural relations among them [24, 25].

We expose ELPIS to audio clips of notes rendered sequentially as ascending major scales within an octave. The recordings retain natural variations in dynamics and articulation, requiring the model to abstract over such variability. Analogous to the number task, we train a VQ-VAE to learn discrete representations of input signals. Here, the contextual structure of symbols is modeled by two learned modules rather than a single reasoning function: one captures long-term sequential symbolic evolution (serving as R), and the other captures local transformations between symbolic states (serving as T). Learning is again guided by intrinsic physical symmetry: whether a major scale is unfolded autoregressively by R before or after applying a transformation T , the resulting symbolic trajectory should remain consistent (Fig. 5A). From a musical perspective, this suggests that the learned representations support *transposition*—the ability to shift pitch patterns consistently across keys [26, 27].

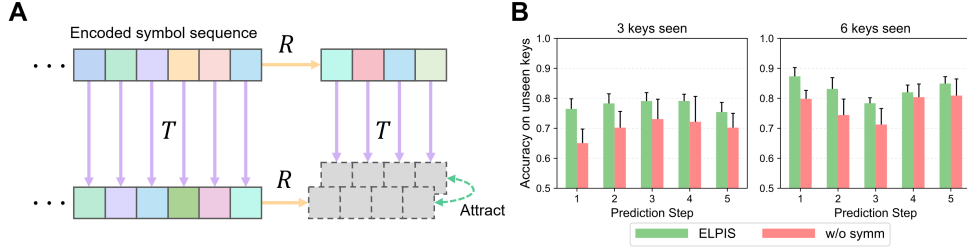


Fig. 5 ELPIS in the emergence of pitches and scales. (A) The intrinsic physical symmetry constraint. For any sequence of symbolic representations encoded from audio clips of notes, the model is constrained with $T(R(\cdot)) = R(T(\cdot))$. (B) Comparison of future-step prediction accuracies in learning music scales based on decoder outputs, evaluated by a pretrained pitch classifier. Two setups are shown: training and validation on (left) 3 keys or (right) 6 keys, with the remaining keys held out for testing. All models are well-trained and perform comparably on the validation set.

We again find that ELPIS learns symbolic language of pitch and its temporal dynamics on scales in different keys, demonstrating generalization beyond memorization: it can infer the continuation of scales it has never encountered. Fig. 5B shows how models trained on a subset of keys generalize to the remaining, unseen keys, using 5-step-ahead future-prediction accuracy measured from decoded audio with a pretrained pitch classifier. Whether trained on 3 or 6 of the 12 keys, ELPIS (green bars) consistently outperforms the baseline (brown bars) in 5-step-ahead prediction accuracy on unseen keys, with an average absolute gain of 7%. Importantly, all models perform equally well on the seen keys, indicating that the improvement reflects a genuine shift from memorization to generalization. Again, transposition is not imposed a priori; instead, the model learns from data a local transformation T corresponding to transposition by 1 or 2 semitones (see Supplementary Section S.2.5).

3 Discussion

In this work, we show that symbolic structure, such as natural numbers and addition, can emerge directly from raw perceptual input via intrinsic physical symmetry, without predefined labels or even transformation groups. The same mechanism also uncovers musical pitch and scale structure from raw audio, illustrating cross-domain generality. Taken together, these results suggest that physical symmetry provides a structural pathway through which symbolic reasoning (System 2) can arise from perceptual learning (System 1).

This organizing role of symmetry resonates with the development of modern deep learning systems. Classical architectures such as convolutional neural networks internalize equivariance between inputs and representations [28–30]. More recent self-supervised approaches—including Siamese and contrastive methods—exploit invariance under input transformations for learning [19, 31–37]. Further work has begun to impose symmetry directly on latent representations and their dynamics, yielding interpretable low-dimensional structures [15]. However, in all such cases, the transformation groups (e.g., translation or rotation) are externally specified. Our work takes

a step further: rather than assuming a certain symmetry, ELPIS learns transformations from data. In this sense, symmetry is not merely a modeling constraint, but a meta-level organizing principle of learned representations.

This work is also related to the broad domain of concept discovery in modern learning systems. It has been well studied that large pretrained models already contain and imply concepts [38–40]—e.g., concepts such as number or pitch can often be recovered via probing [41], and some models may even use these concepts implicitly when making predictions [42–45]. However, probing presupposes knowing what to look for and requires labeled examples to identify the desired concepts in the first place. In other words, a concept becomes explicitly usable only if an external agent (typically a human) specifies what the labels are and how they should be read out. Even recent studies that automatically extract concepts from pretrained latent spaces still depend on an external observer to interpret the factors [46–48]. In contrast, our approach treats concepts not as implicit structures to be extracted post hoc, but as explicit outcomes of learning that can be directly operated upon.

Despite these results, several important directions remain open. First, natural numbers and addition represent only the simplest form of symbolic mathematical language. ELPIS does not yet acquire more advanced numerical systems, such as positional representations that enable unbounded compositional generalization [49]. Similarly, in music, pitch can be organized through richer structures such as octave and pitch classes [24, 26]. The emergence of such higher-order structures may require additional mechanisms, potentially including multi-agent interaction. Second, intrinsic physical symmetry has so far been studied in settings involving a single class of concepts. Whether the same principle extends to multiple interacting concept types—such as objects, relations, and actions—remains an open question. Together, these directions point toward a broader question: whether symmetry can serve as a general principle for the emergence of structured, compositional languages across domains—and, in doing so, bring emergent machine languages closer to the expressiveness of human language [50–52].

4 Methods

4.1 Number and addition datasets

The main number experiment used visual triplets (x_m, x_n, x_p) representing ordered addition relations $m + n = p$, with $0 \leq m, n, p \leq 20$. Relations whose sums exceeded 20 were excluded. The primary dataset consisted of deterministic blue-dot arrays: each integer $k \in \{0, \dots, 20\}$ was rendered as a single fixed 64×64 RGB prototype, with a hollow circle used for zero. Each training or test sample contained the two addend images and the corresponding sum image. Because commutativity was not assumed, (m, n) and (n, m) were treated as different ordered relations. The fixed split used for the main results (Fig. 2A and Fig. 3) contained 68 seen addend pairs and 163 held-out pairs; the full split is listed in Supplementary Table S1. For the sample-efficiency analysis (Fig. 2B), the train–test split was randomly resampled at each training ratio.

We additionally tested visual variants to examine whether the learned symbols depended on the exact appearance of the blue-dot prototypes. Eastern and Western tally marks used the same addition relations and the same range $0, \dots, 20$, but replaced dot arrays with structured stroke-based prototypes. Random blurred dots applied Gaussian blur online to each image, with kernel size sampled from $\{5, 7, 9\}$ and standard deviation $\sigma \sim \mathcal{U}(0.5, 3.0)$. Multi-style dots associated each number with 16 visual realizations generated from four colors and four shapes. In these multiple-realization settings, evaluation was performed over repeated visual instantiations and code assignments were mapped back to integer labels by majority vote when a single number could occupy multiple discrete codes. Results for these variants are reported in Supplementary Section S.1.7.

4.2 ELPIS model for visual addition

ELPIS combines a vector-quantized autoencoder with a learned reasoning function over discrete latent codes. Given an input image x , a CNN encoder followed by an MLP maps the image to a continuous latent vector $z = f_\varphi(x)$. Vector quantization then assigns the latent to a learned codebook $\mathcal{C}$, giving the symbolic encoder

$$f_\phi(x) = Q_{\mathcal{C}}(f_\varphi(x)), \quad \phi = (\varphi, \mathcal{C}). \quad (1)$$

In the main addition experiments, the latent vector was divided into $L = 2$ parts, each quantized with a shared codebook containing $M = 10$ entries of dimension $d_s = 1$. This gives a symbolic capacity of $K = M^L = 100$ possible codes, with each quantized embedding $e \in \mathbb{R}^{Ld_s}$. A decoder g_ψ , implemented as an MLP followed by transposed convolutional layers, reconstructs the input from the quantized code.

The reasoning function h_θ is a learned binary operator on symbolic codes. Given codes e_m and e_n , it predicts a code for their composed result:

$$e_{m \oplus n} = h_\theta(e_m, e_n) = Q_{\mathcal{C}}(\text{MLP}_\theta([e_m, e_n])), \quad (2)$$

where $[\cdot, \cdot]$ denotes concatenation and $\oplus$ denotes the operation learned by the model rather than a predefined arithmetic operator. The same discrete code space is therefore used for perception, reconstruction, and symbolic reasoning.

4.3 Intrinsic physical symmetry for addition

Physical symmetry can be expressed as an equivariance relation between a law R and a transformation T , namely $T(R(\cdot)) = R(T(\cdot))$. In the addition setting, both R and T are instantiated by the learned reasoning function h_θ itself. For arbitrary symbolic codes a, b, c , this yields path-equivalence constraints over different ways of composing the same learned operator. Additional derivations and alternative parameterizations are provided in Supplementary Section S.1.2. The main experiments used the two ordering-consistent forms

$$h_\theta(h_\theta(a, b), c) = h_\theta(h_\theta(a, c), b), \quad (3)$$

and

$$h_\theta(c, h_\theta(b, a)) = h_\theta(b, h_\theta(c, a)). \quad (4)$$

These constraints do not assume that the operation is commutative, nor do they impose a predefined group action. Instead, they regularize the learned latent operation so that alternative computational paths agree.

For a triplet of encoded symbolic codes, we implement this principle as an embedding-space path-equivalence loss,

$$\mathcal{L}_{\text{sym}}^1 = \|e_{(a \oplus b) \oplus c} - e_{(a \oplus c) \oplus b}\|_2^2, \quad (5)$$

$$\mathcal{L}_{\text{sym}}^2 = \|e_{a \oplus (b \oplus c)} - e_{b \oplus (a \oplus c)}\|_2^2, \quad (6)$$

with $\mathcal{L}_{\text{sym}} = \mathcal{L}_{\text{sym}}^1 + \mathcal{L}_{\text{sym}}^2$.

4.4 Training objective for visual addition

The visual addition model is trained end-to-end from image triplets. For each triplet (x_m, x_n, x_p) , all three images are encoded and reconstructed, giving the reconstruction loss

$$\mathcal{L}_{\text{rec}} = \frac{1}{3} \sum_{u \in \{m, n, p\}} \|g_\psi(e_u) - x_u\|_2^2. \quad (7)$$

The predicted composed code $e_{m \oplus n} = h_\theta(e_m, e_n)$ is constrained to match the sum image and the encoded sum code through an image-level and latent-level addition loss:

$$\mathcal{L}_{\text{oper-img}} = \|g_\psi(e_{m \oplus n}) - x_p\|_2^2, \quad \mathcal{L}_{\text{oper-z}} = \|e_{m \oplus n} - e_p\|_2^2. \quad (8)$$

For vector quantization, we use

$$\mathcal{L}_{\text{VQ}}(e, z) = \lambda_{\text{emb}} \|e - \text{sg}[z]\|_2^2 + \lambda_{\text{com}} \|z - \text{sg}[e]\|_2^2, \quad (9)$$

where $\text{sg}[\cdot]$ denotes stop-gradient. We use $\lambda_{\text{emb}} = 0.01$ and $\lambda_{\text{com}} = 0.0025$, and apply this quantization loss to both encoder outputs and reasoning-function outputs. The complete loss is

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \mathcal{L}_{\text{VQ}}^{\text{enc}} + \lambda_{\text{img}} \mathcal{L}_{\text{oper-img}} + \lambda_z \mathcal{L}_{\text{oper-z}} + \lambda_{\text{sym}} \mathcal{L}_{\text{sym}} + \lambda_{\text{VQ+}} \mathcal{L}_{\text{VQ}}^{\text{op}}, \quad (10)$$

with $\lambda_{\text{img}} = 3.0$, $\lambda_z = 0.02$, $\lambda_{\text{sym}} = 0.02$, and $\lambda_{\text{VQ+}} = 0.5$. Models were optimized with Adam using a learning rate of 10^{-4} , batch size 128, and 50,000 training epochs. Because the main fixed split contains only 68 ordered training pairs, an epoch corresponds to a small number of optimization updates; the schedule therefore exposes each model to many repeated passes through a deliberately limited relation set. Unless otherwise stated, each configuration was evaluated over 20 independent random initializations.

4.5 Baselines, ablations, and repeated runs

All visual-addition comparisons in the main number experiments used the same symbolic capacity and optimizer settings unless otherwise stated. The full ELPIS model used the complete objective above. The model without symmetry used the same encoder, decoder, reasoning function, and operation losses, but removed the intrinsic-symmetry term by setting $\lambda_{\text{sym}} = 0$. The reconstruction-only baseline removed the reasoning module and all operation-level and symmetry losses, leaving only the VQ-VAE reconstruction branch; this baseline tests whether discrete perceptual codes alone yield usable numerical structure. The ceiling model used the same architecture and no-symmetry objective as the model without symmetry, but was trained on all ordered addend pairs rather than on the partial split. The random baseline used randomly assigned symbolic representations for representational analyses such as orderliness and subtraction transfer. In the music experiments, the baseline used the same autoencoding, unfolding, and transition modules as ELPIS but removed the symmetry term $\mathcal{L}_{\text{symm}}$.

For the fixed-split number experiments, summary statistics were computed over 20 independent runs with different random initializations. For the sample-efficiency analysis (Fig. 2B), the train-test split was resampled for each run at each training ratio. For subtraction transfer (Fig. 3A), the frozen representations from each independently trained addition model were evaluated with independently initialized subtraction heads.

4.6 Evaluation of learned numerical representations

The primary generalization metric is addition accuracy on held-out ordered addend pairs. Let $\mathcal{T}_{\text{add}}$ denote this held-out set. In hard matching, a query (m, n) is correct only if the predicted code is quantized to exactly the same discrete code as the encoded image of the true sum $p = m + n$:

$$\text{Acc}_{\text{add}}^{\text{hard}} = \frac{1}{|\mathcal{T}_{\text{add}}|} \sum_{(m,n) \in \mathcal{T}_{\text{add}}} \mathbb{1}[e_{m \oplus n} = e_p]. \quad (11)$$

We also report a distance-based criterion in which the predicted code is assigned to the nearest integer prototype in the learned latent space. This provides a complementary accuracy measure and does not change the qualitative conclusions.

To test whether the learned numerical space supports non-integer semantics, we constructed midpoint codes $e_{k+0.5} = (e_k + e_{k+1})/2$ and evaluated addition queries involving paired half-step shifts whose integer sum remained unchanged. The prediction was scored by the same hard-matching and distance-based criteria used for integer addition. We quantified the spatial structure of the learned number codes using orderliness [18], which asks whether numerically adjacent symbols are also nearest neighbors in the latent space. Finally, to test transfer to a related operation, we froze the encoder and codebook learned from addition and trained a lightweight supervised subtraction

head on 152 subtraction pairs, evaluating it on 79 held-out pairs. Detailed formal definitions for the complementary distance-based criterion, interpolation, orderliness, and subtraction-transfer analyses are provided in Supplementary Section S.1.5.

4.7 Perceptual adaptation to complex icons

To test whether number concepts learned from simple dot arrays could be reused for realistic visual scenes (Fig. 4), we froze a pretrained ELPIS addition model and learned only a perceptual adapter. We first extracted prototype number codes from the frozen model and restricted the adaptation dictionary to the ten codes $\mathcal{C}' = \{e_1^*, \dots, e_{10}^*\}$, corresponding to numbers 1–10. The frozen reasoning function was then used with this restricted dictionary:

$$h_{\theta'}(e_m, e_n) = Q_{\mathcal{C}'}(\text{MLP}_{\vartheta}([e_m, e_n])), \quad \theta' = (\vartheta, \mathcal{C}'). \quad (12)$$

Here, ϑ and $\mathcal{C}'$ were fixed during adaptation.

Icon images were rendered on 224×224 RGB canvases with randomized object sizes and non-overlapping positions. A frozen DINO ViT-S/8 encoder [19] produced a 384-dimensional feature vector for each image. A trainable adapter v_ω , a three-hidden-layer MLP with widths [1024, 1024, 1024], LeakyReLU activations with negative slope 0.01, and no normalization, mapped DINO features into the frozen ELPIS number-code space. Only v_ω was trained, using Adam with learning rate 10^{-4} .

Training triplets were generated online by recursively partitioning icon sets. Starting from ten non-overlapping objects, we sampled a split $p = m + n$, rendered the full image x_p , and rendered two complementary subsets x_m and x_n . The larger subset was then recursively split to produce additional self-supervised triplets from the same scene. During training, adapter outputs were quantized to $\mathcal{C}'$, and the frozen reasoning function was required to map the two subset codes to the code of the full image. The adaptation loss combined VQ losses for adapter and reasoning outputs, an addition consistency loss with stop-gradient on alternating branches, and a collapse-avoidance term that penalized triplets whose codes became less separated than the minimum pairwise distance among the fixed prototype codes. Both the addition-consistency and collapse-loss weights were set to 1.

We evaluated counting accuracy by quantizing adapted icon codes and mapping each code to its corresponding number in the restricted dictionary. In-distribution tests used the ten icon types seen during adaptation, while out-of-distribution tests used five unseen icon types. The adaptation procedure occasionally converged to collapsed alignments. After inspecting training-set counting accuracies, we observed a clear separation between collapsed runs, all of which had training accuracy ≤ 0.40 , and converged runs, all of which had training accuracy ≥ 0.97 . We therefore used a training-stage convergence criterion requiring average training counting accuracy greater than 0.40. This criterion was based only on training-set performance and did not use test-set accuracy. Ten of 20 independent adaptation trials satisfied this criterion; main-text icon results are reported conditional on these converged trials, and the full set of adaptation trials is reported in Supplementary Section S.1.8 and Supplementary Table S2.

4.8 Music-scale experiments

For the music experiment (Fig. 5), ELPIS was trained on mel-spectrogram segments of monophonic notes rendered from a saxophone synthesizer playing ascending major scales. The dataset contained 12 major keys within a single octave, MIDI pitches 48–59; when a scale exceeded the octave, it wrapped to the lowest note. Each note lasted one second, with velocity sampled between 80 and 120 and randomized amplitude envelopes. Sequences were extended to 90 notes. Audio was synthesized at 16 kHz, normalized, converted to magnitude spectrograms with FFT size 1024, window length 1024, and hop size 256, and then mapped to 128-bin mel-spectrograms. In the three-key setting, C, E, and G# were used for training and validation; in the six-key setting, C, G, D, A, E, and B were used for training and validation. The remaining keys were held out for testing.

The music model used a ResNet-like convolutional sequence autoencoder with a VQ bottleneck. The encoder mapped each spectrogram segment to a $D = 128$ -dimensional latent vector, and the decoder mirrored the encoder with transpose-convolutional stages. Quantization used a codebook with $N_{\text{atoms}} = 24$ entries, allowing redundancy beyond the 12 pitch classes. The contextual structure of the learned symbols was modeled by two modules: an autoregressive unfolding module h_{ζ} , implemented as a single-layer RNN with hidden size D , and a transition module k_{η} , implemented as a two-layer energy network that assigns transition probabilities between codebook entries.

The music objective combined reconstruction, VQ commitment, sequential dynamics, transition modeling, and intrinsic symmetry. The unfolding module predicted future symbols from previous symbols using mean squared error, with teacher forcing ratio 0.9. The transition module was trained to assign high probability to adjacent symbolic transitions in encoded sequences. Intrinsic symmetry required that continuing a scale and then applying a learned transition produced the same latent trajectory as applying the transition first and then continuing the scale:

$$\mathcal{L}_{\text{symm}} = \|T(R(\mathbf{e})) - R(T(\mathbf{e}))\|_2^2. \quad (13)$$

The full objective was

$$\mathcal{L}_{\text{ELPIS}} = \mathcal{L}_{\text{recon}} + \beta \mathcal{L}_{\text{commit}} + \gamma \mathcal{L}_{\text{unfolding}} + \rho \mathcal{L}_{\text{transition}} + \lambda \mathcal{L}_{\text{symm}}, \quad (14)$$

with $\beta = \gamma = \rho = 1$ and $\lambda = 0.2$. VQ codebooks were updated using exponential moving averages, and the rotation trick was used for gradient estimation through quantization [53]. For each setting, five models were trained for 6,000 steps with batch size 256 using AdamW at learning rate 10^{-4} and cosine annealing every 2,000 steps. Training used a single NVIDIA RTX 5000 Ada GPU, and checkpoints were selected by validation loss.

Music generalization was evaluated as future-symbol prediction from a seven-note prompt. Predicted symbols were decoded back to spectrograms and classified by a separately trained pitch classifier over 12 pitch classes. This classifier used the lowest 64 mel bins, two convolutional layers with 32 and 64 channels, ReLU activations, dropout,

global max-pooling over time, and a small MLP classifier. A decoded prediction was counted as correct when the classifier’s pitch label matched the ground truth. We also evaluated predictions directly in symbolic space by assigning each codebook entry to the pitch class it most frequently encoded in the test set and scoring whether predicted symbols matched the ground-truth pitch labels. Additional music dataset, architecture, training, symbolic-space evaluation, and transition analyses are provided in Supplementary Sections S.2.1–S.2.5.

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Declarations

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Data, code, and materials availability. Code and datasets for reproducing the analyses are available at <https://github.com/XuanjieLiu/ELPIS>.

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